§11.1 Sequences

Homework：13,19,32,33,51,61

Definition：

- A sequence is a function whose domain is N=the set of positive integers.
- In notation, we write $f(n)=a_n$.

Question：Convergence or divergence of $\{a_n\}$.

Example 1：

i. $a_n=\dfrac{1}{n}$ convergence

ii. $a_n=(-1)^n$ divergence

iii. $a_n=n$ divergence

Theorem1： $\lim\limits_{x\to\infty} f(x)=L$, $x\in R$ $\Rightarrow \lim\limits_{n\to\infty} f(n)=L$, $n\in N$.

(The converse is not true.)

Theorem2： $\lim\limits_{n\to\infty}|a_n|=0$, then $\lim\limits_{n\to\infty} a_n=0$

Theorem3： $\{r^n\}_{n=1}^{\infty}$ converges iff $-1<r\le 1$.

Example 2： $a_n=\dfrac{\ln n}{n}$

$$\lim_{n\to\infty}\frac{\ln n}{n}=\lim_{n\to\infty}\frac{\frac{1}{n}}{1}=0 \qquad \text{(X)}$$

$$\lim_{x\to\infty}\frac{\ln x}{x}=\lim_{x\to\infty}\frac{\frac{1}{x}}{1}=0+\text{Theorem1}\Rightarrow\lim_{n\to\infty}\frac{\ln n}{n}=0 \qquad \text{(O)}$$

註：

i. 在離散點上講微分是無意義的.

ii. $n^n >> n! >> e^n > n^3 > \ln n$ when n large.

Theorem4： Let $\{a_n\}$ be monotonic(單調) and bounded(有界) $\Rightarrow \{a_n\}$ converges.

Example 3： Prove that $\sqrt{2+\sqrt{2+\sqrt{2+..}}}$ exists.

Proof： Let $a_1 = \sqrt{2}$, $a_{n+1} = \sqrt{2+a_n}$

(i) Clearly, a_n is monotonic increasing.

(ii) Moreover, $a_n \le 2$ for all $n \in N$.

We prove (ii) by induction.

(a) $a_1 \le 2$

(b) Suppose $a_n \le 2$, Then $a_{n+1} = \sqrt{2+a_n} \le \sqrt{2+2} = 2$

Using Theorem4, we see that $\lim_{n\to\infty} a_n$ has a limit, say x.

Then it must satisfy

$$\lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \sqrt{2+a_n}$$

That is

$$x = \sqrt{2+x} \Rightarrow x = 2 \text{ or } -1(\text{不合})$$